nuclides 1, 2, etc., the individual cross-sections are $\sigma_1$, $\sigma_2$, etc. The macroscopic cross-section $\hat{\sum}(m^{-1})$ is

$$\hat{\sum} = N_v \sum_1^n \gamma_i \sigma_i \tag{11.6}$$

For a target which is $x$ m thick one obtains

$$\varphi = \varphi_o e^{-\hat{\sum} x} \tag{11.7}$$

The value $\hat{\sum}^{-1}$ is the average distance a projectile travels between successive collisions with the target atoms (the *mean free path*).

## 11.2. PARTIAL REACTION CROSS-SECTIONS

The irradiation of a target may lead to the formation of a number of different products. For example, the irradiation of $^{63}$Cu with protons can produce the nuclides $^{62}$Zn, $^{63}$Zn, $^{62}$Cu, all of which are radioactive:

$$^{63}\text{Cu} + {}^{1}\text{H} \rightarrow \begin{cases} (\text{p,n}) & {}^{63}\text{Zn} + \text{n} & (11.8a) \\ (\text{p,2n}) & {}^{62}\text{Zn} + 2\text{n} & (11.8b) \\ (\text{p,pn}) & {}^{62}\text{Cu} + {}^{1}\text{H} + \text{n} & (11.8c) \end{cases}$$

The formation probability of each product corresponds to a *partial reaction cross-section*. The total reaction cross-section $\sigma_{tot}$ is the sum of all the partial cross-sections and measures the probability that the projectile causes a nuclear reaction independent of the products formed. Thus, the decrease in intensity of the particle flux is proportional to $\sigma_{tot}$. The amount of an individual product formed is proportional to $\sigma_i$, where $\sigma_i$ corresponds to the partial reaction cross-section for the formation of the $i$:th product.

Some partial cross-sections have their own names such as the *scattering cross-section* for elastic and inelastic scattering ($\sigma_{scat}$), the *activation cross-section* ($\sigma_{act}$) for the formation of radioactive products, the *fission cross-section* ($\sigma_f$) for fission processes, and *adsorption* or *capture cross-sections* ($\sigma_{abs}$ or $\sigma_{capt}$) for the absorption or capture of particles. If all of these processes take place, one obtains (with caution to avoid overlapping reactions)

$$\sigma_{tot} = \sigma_{scat} + \sigma_{act} + \sigma_f, \text{ etc} \tag{11.9}$$

In the irradiation of $^{235}$U with thermal neutrons, $\sigma_{scat}$ is about 10 b, $\sigma_{act}$ (for forming $^{236}$U) is approximately 107 b, and $\sigma_f$ is 582 b.

The reaction cross-section depends on the projectile energy as shown in Figure 11.2. The curves obtained for the partial reaction cross-section as a function of projectile energy are known as *excitation functions* or *excitation curves*.

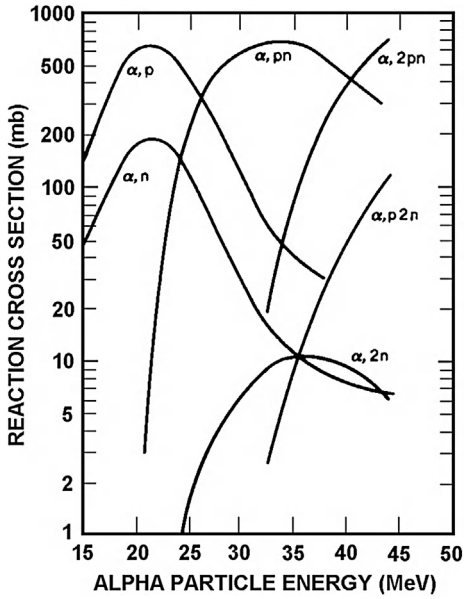



**Figure 11.2** Excitation functions for reactions between $^{4}$He ions and $^{54}$Fe target nuclei. The kinetic energy of the projectile is in the laboratory system. (From Houck and Miller.)

## 11.3. RESONANCE AND TUNNELING

Experimentally, it is found that nuclear reactions sometimes occur at energies less than that required by the Coulomb barrier. This behavior is related to the wave mechanical nature of the particles involved in a nuclear reaction.

As a projectile approaches a target nucleus in a nuclear reaction, the probability that there will be overlap and hence interaction in their wave functions increases. This concept was used in §6.7.3 to explain the emission of α-particles with energies less than that required to overcome the Coulomb barrier. Such tunneling may also occur for projectiles approaching the nucleus from the outside. An example is provided by the reaction of protons with lithium (Fig. 11.3).

$$^{7}_{3}\mathrm{Li} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{8}_{4}\mathrm{Be}^{*} \rightarrow \begin{cases} 2\,{}^{4}_{2}\mathrm{He} & Q = 17.4\ \mathrm{MeV} \quad (11.10a) \\ {}^{8}_{4}\mathrm{Be} + \gamma & Q = 17.2\ \mathrm{MeV} \quad (11.10b) \\ {}^{7}_{4}\mathrm{Be} + \mathrm{n} & Q = -1.64\ \mathrm{MeV} \quad (11.10c) \end{cases}$$

For this reaction the value of $E_{\mathrm{cb(min)}}$ is 1.3 MeV. However, due to tunneling the reactions begin to occur at lower proton energies. At an energy of 0.15 MeV about 0.1% of the protons penetrate the Coulomb barrier, at 0.3 MeV about 1%, and at 0.6 MeV about 20%.